Does Posterior Uncertainty Widen Across an Open Regularity Interval? A Dyadic Shell Model Test.

Grace Dice

***Abstract*—This paper examines whether the posterior uncertainty of a Bayesian classifier widens across a provably open regime. The testbed is a dyadic shell model of the Navier-Stokes equations whose damping exponent α has a proven partition: finite-time blow-up for $\alpha < 1/3$, global regularity for $\alpha \geq 1/2$, and an open interval between. A Bayesian logistic classifier, sampled with a Metropolis-Hastings sampler implemented from scratch, was trained on resolution-sweep features from the two proven zones only. Mean credible-interval width in the open interval was 8–13$\times$ that in the proven zones, with small variation across five random seeds. Held-out accuracy was 100%, including near both proven boundaries. The width peak ($\alpha \approx 0.37$–0.38) did not coincide with the feature-sensitivity peak ($\alpha \approx 0.25$–0.30), which weakens one artifact explanation. The widening may also reflect the classifier's decision boundary, and no control system without an open problem was run. The result is exploratory and does not address the open regularity problem.**

***Index Terms*—Bayesian inference, epistemic uncertainty, shell models, dyadic models, uncertainty quantification**

I. Motivation

A Bayesian credible interval claims to measure how strongly the data constrain a prediction. That claim is rarely checked against external ground truth, which is usually unavailable. Standard calibration diagnostics typically test only internal consistency: whether stated confidence matches empirical accuracy on data from the training distribution.

The Desnyansky-Novikov dyadic shell model permits a stronger test. For its damping exponent α, finite-time blow-up is proven for $\alpha < 1/3$, local regularity for $\alpha > 1/3$, and global regularity for all time once $\alpha \geq 1/2$ [1]. For $1/3 \leq \alpha < 1/2$, whether local solutions remain smooth for all time or eventually blow up is open. Provable answers therefore flank a regime with no known answer.

If posterior uncertainty reflects limits on the information in the numerical features, rather than interpolation between labelled classes, it should rise in the open interval. This paper builds a resolution-sweep feature extractor and a Metropolis-Hastings sampler over a Bayesian logistic model, trains on the proven zones only, and measures credible-interval width across α. Width is larger in the open interval, peaks nearer the blow-up boundary, and is small on both proven sides, including immediately adjacent to the thresholds. This is consistent with the hypothesis but does not establish it (Section VI).

II. Related Work

Senge et al. proposed a classifier that separates aleatoric from epistemic uncertainty [2], and recent work examines where common uncertainty-quantification methods fail to separate them [3]. Neither work, as far as found, considers a system where the correct answer is provably unknown rather than merely unobserved.

Bayesian methods have been applied to Navier-Stokes-adjacent problems at a different level. Calibration of RANS turbulence-model parameters treats model-form discrepancy as the object of inference [4], and Bayesian differential-equation solvers propagate numerical uncertainty through Navier-Stokes simulations [5]. Neither, as far as found, tests whether posterior uncertainty tracks a region of unresolved mathematics.

Palasek constructed finite-time blow-up in an elementary shell model of the 3D Navier-Stokes equations [6], within the dyadic-model literature that supplies the forward model here. No prior study of the present combination was found. This paper is exploratory and makes no novelty claim against the full literature.

III. Background

The dyadic shell model is a low-dimensional system intended to mimic the Navier-Stokes energy cascade without a spatial PDE. The scalar $\psi_n(t)$ is the energy at length scale λ^{-n} for shells $n = 0, 1, 2, \ldots$, and it evolves as

$$\partial_t \psi_n = -\nu \lambda^{2n\alpha} \psi_n + \lambda^n \psi_{n-1}^2 - \lambda^{n+1} \psi_n \psi_{n+1}. \qquad (1)$$

Here $\nu > 0$ is a fixed viscosity, $\lambda > 1$ is the shell spacing ratio ($\lambda = 2$ throughout), and α sets the strength of small-scale dissipation. Simulations truncate the system at N shells.

The first term damps shell n at a rate that grows with n. The second and third terms couple adjacent shells: energy enters shell n from shell $n-1$ and leaves toward shell $n+1$. In the intended picture, energy cascades from large to small scales until viscosity removes it.

Shell models retain this cascade while discarding spatial structure, which makes rigorous results possible. They are studied because finite-time singularity formation in the 3D Navier-Stokes equations remains an open problem.

Here the shell model serves only as a testbed: it is simple enough to simulate at low cost and has a known ground-truth partition of its parameter space. No claim is made about the open interval or the full Navier-Stokes equations.

IV. Methods

A. Forward Model

The model was implemented as coupled ODEs (shell_rhs) with an analytical tridiagonal Jacobian (shell_jac). The Jacobian served implicit integration and a Newton solve in a second, independently coded time-stepping scheme. The tracked scalar was a Sobolev-norm diagnostic: a weighted sum over shells with weight exponent $s = 0.5$.

B. Solver Validation

To check for solver artifact, MATLAB ODE15s (variable-order BDF) was compared against a from-scratch fixed-step Crank-Nicolson scheme with Newton iteration, at $\alpha = 0.2$ (blow-up regime) and $N = 40$. The Sobolev-norm trajectories agreed closely (Fig. 1). This is consistent with the growth at small α reflecting the dynamics rather than solver stiffness or truncation error. Agreement at a single α and N does not exclude artifacts elsewhere.

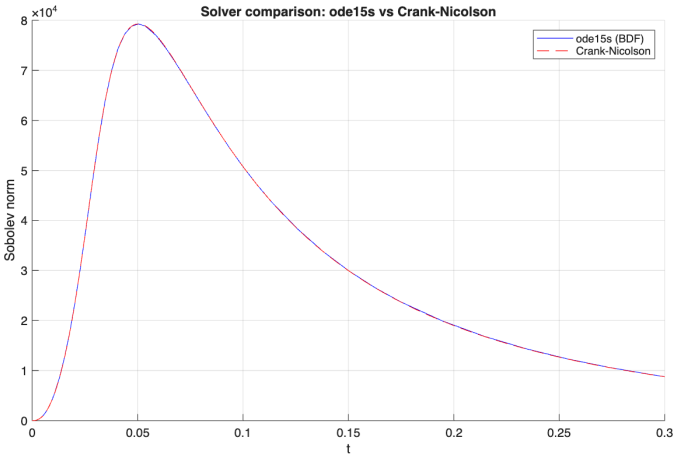


Fig. 1. Sobolev norm versus time at $\alpha = 0.2$, $N = 40$, computed with ODE15s (BDF) and a fixed-step Crank-Nicolson/Newton scheme. The two trajectories agree, consistent with the growth being a property of the model rather than of the solver.

C. Feature Extraction

For each α, the model was run at $N = 15, 20, 25, 30, 40$, and the peak Sobolev norm, the time of that peak, and the growth rate of the norm were recorded. A norm still rising at the end of the window yields a spuriously low peak, so the extraction routine (run_and_extract) extended the integration time whenever the peak fell on the last time sample, up to a fixed cap.

Five features were computed per α from the resolution sweep: (i) the slope of log-peak versus N, (ii) the drift in peak time across resolutions, (iii) the residual spread about the slope fit, (iv) the peak value at the highest resolution, and (v) the maximum growth rate at the highest resolution. A steep positive slope indicates a peak that keeps growing with resolution, which is characteristic of blow-up. A near-zero slope indicates a converged peak, which is characteristic of bounded behaviour.

D. Labelling

A grid of α values spanning all three zones was run with several random initial-condition perturbations per α. Rows with $\alpha < 1/3$ were labelled "blowup," rows with $\alpha \geq 1/2$ "regular," and rows with $1/3 \leq \alpha < 1/2$ "unknown." Labels therefore follow the theoretical zone, not the observed behaviour of individual runs. Only labelled rows entered the fit. Unknown rows were passed through the trained model for posterior predictions.

E. Bayesian Classifier

A logistic regression on the five features used a Gaussian prior on each coefficient. Inference used a random-walk Metropolis-Hastings sampler written from scratch: a Gaussian perturbation of the current coefficient vector was proposed and accepted according to the log-posterior ratio. Features were standardized with the mean and standard deviation of the labelled rows only, which avoids leakage from the open zone. For any α, the posterior samples give a distribution over the probability of blow-up.

F. Diagnostics

Five checks were applied: acceptance rate and trace plots; open-zone widening across five independent random seeds; a held-out calibration, training on a subset of initial-condition replicates and testing on the remainder (Brier score and accuracy); re-sampling densified near both proven boundaries; and multi-chain Gelman-Rubin (R-hat) and effective sample size on the final model.

V. Results

A. Open-Zone Credible Intervals

The mean credible-interval width was 0.047 in the blow-up zone, 0.380 in the open zone, and 0.029 in the regular zone (Fig. 2). The open zone was thus roughly $8\times$ wider than the blow-up zone and $13\times$ wider than the regular zone. Across five independent random seeds, open/blowup ratios ranged from 7.92 to 8.43 (mean 8.15, std 0.18) and open/regular ratios from 12.26 to 13.41 (mean 12.95, std 0.50). The small spread indicates that the widening is not attributable to sampling noise in a single chain.

B. Structure Within the Open Zone

Widening is not uniform across the open interval (Fig. 4). Width spikes just above $\alpha = 1/3$, reaching a maximum near 0.94 at $\alpha \approx 0.37$–0.38, then decays toward the regular-zone baseline as α approaches $1/2$. The width histogram (Fig. 3) is bimodal, with a cluster near zero and a second spread over 0.4–0.9. Width is therefore concentrated near the blow-up boundary and falls toward the regularity boundary.

C. The tpeak_drift=0 Artifact

An earlier pipeline version recorded zero tpeak_drift for several small-α blow-up rows. Inspection at $\alpha = 0.02, 0.04, 0.06$ for $N = 15$ and $N = 20$ showed the norm still rising at the end of the window in every case (tpeak values of 0.56–0.73, at the edge of the original 0.5-unit window), so the recorded peak was a truncation artifact. After the auto-extension logic was added to run_and_extract, every case resolved to a genuine peak (censored=0, tmax_used extending to 0.5–1.0), and tpeak_drift became uniformly nonzero for $\alpha = 0.02$–0.08. This particular artifact therefore did not enter the training features.

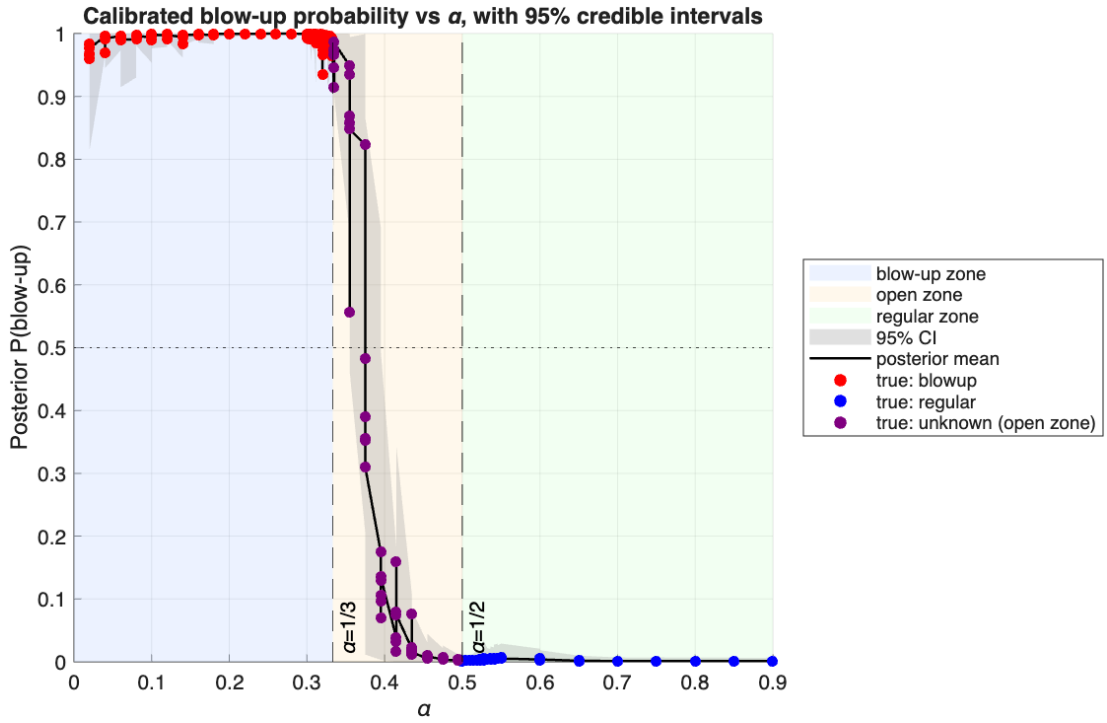


Fig. 2. Posterior mean probability of blow-up (black line) with 95% credible interval (gray band) across the α sweep. Points are colored by ground-truth label; purple points in the open zone ($1/3 \leq \alpha < 1/2$) have no ground truth. The band widens only inside the open zone.

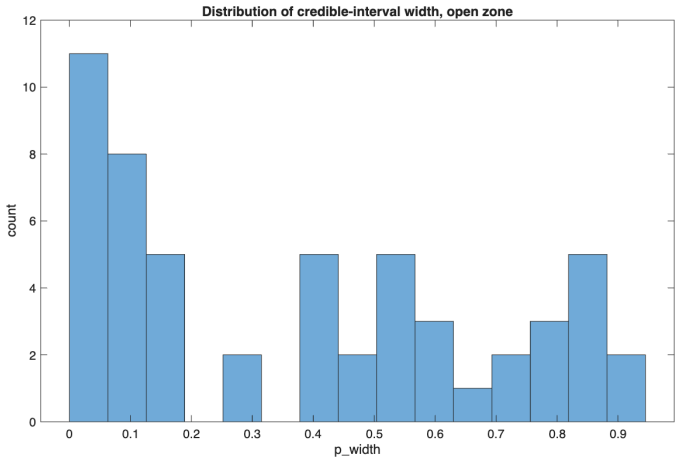


Fig. 3. Distribution of credible-interval width for open-zone points. The distribution is bimodal, with a cluster of narrow intervals and a cluster of wide ones.

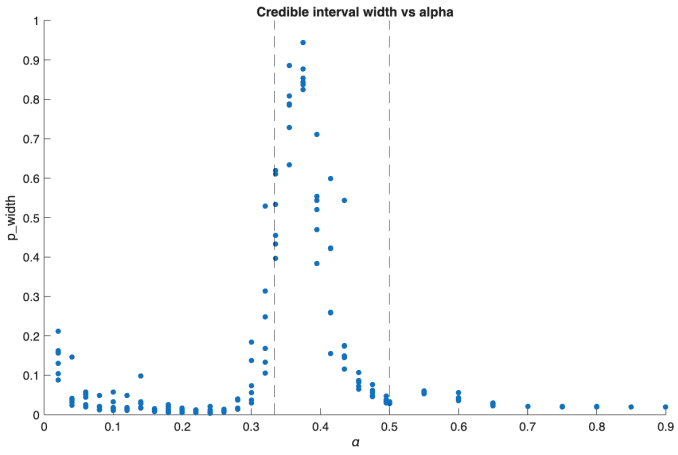


Fig. 4. Credible-interval width versus α. Width spikes just above $\alpha = 1/3$ and decays toward $\alpha = 1/2$.

D. Held-Out and Boundary Calibration

A held-out check, trained on four of six initial-condition replicates and tested on the other two, gave a Brier score of 0.0003 and 100% accuracy. No held-out prediction fell in the ambiguous range $p \in (0.2, 0.8)$, where p is the posterior mean probability of blow-up (Fig. 5). The split is by replicate, so held-out points share α values with training points.

Repeating the procedure on an α grid densified near both proven boundaries gave a Brier score of 0.0001 and 100% accuracy. Predicted blow-up probability stayed at 0.97–1.00 within 0.001–0.01 of $\alpha = 1/3$ and fell to 0.002–0.008 within the same margin of $\alpha = 1/2$. Wide intervals appeared only inside the open interval.

E. Sampler Diagnostics

Acceptance rates across all runs were 0.34–0.38, within the commonly cited target range for random-walk Metropolis-Hastings. Trace plots for all six parameters (intercept and five feature coefficients) were stationary and well mixed (Fig. 6). On the final model, R-hat was 1.0000–1.0005 for all six

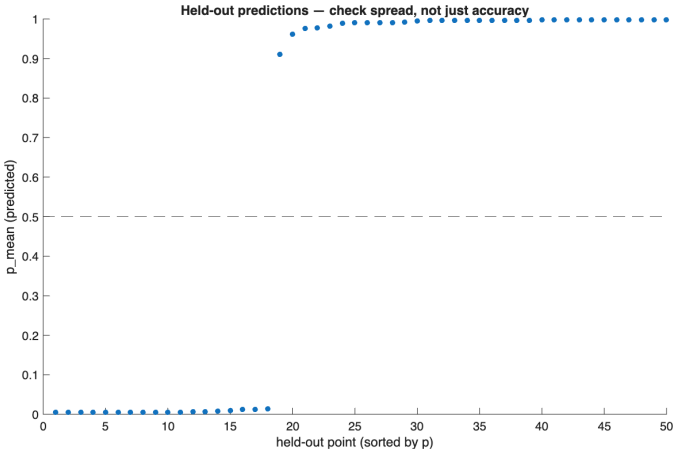


Fig. 5. Posterior mean predicted probability of blow-up for held-out initial-condition replicates, sorted by value. Every point lies near 0 or 1; none falls in the ambiguous range $(0.2, 0.8)$.

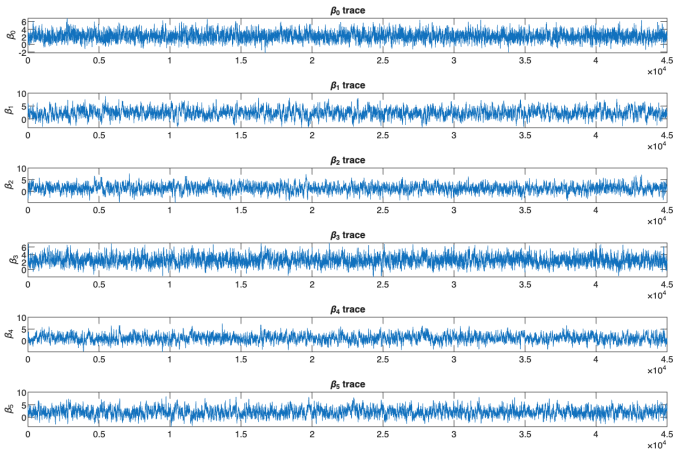


Fig. 6. Trace plots for all six logistic-regression coefficients over 45,000 post-burn-in samples. The chains are stationary and well mixed.

parameters, and effective sample size ranged from 3727 to 7228 out of 45,000 post-burn-in draws.

VI. Discussion

Posterior uncertainty widened inside the interval that theory identifies as unresolved and was small on both proven sides, even immediately adjacent to the thresholds. This matches the hypothesis, but it is also what interpolation between two labelled clusters would produce, so width alone does not distinguish the two explanations.

One alternative explanation is a feature-geometry artifact: if the features change fastest mid-range, a logistic classifier reports maximal uncertainty there regardless of any open problem. Figs. 7–9 test this. The mean slope of log-peak versus N varies smoothly and monotonically across the open interval (Fig. 7). Its sensitivity $|d(\text{slope})/d\alpha|$ peaks near $\alpha \approx 0.25$–0.30 (Fig. 8), earlier than the width peak at $\alpha \approx 0.37$–0.38. A purely inherited widening would produce coincident peaks. The observed offset, with the width peak nearer the

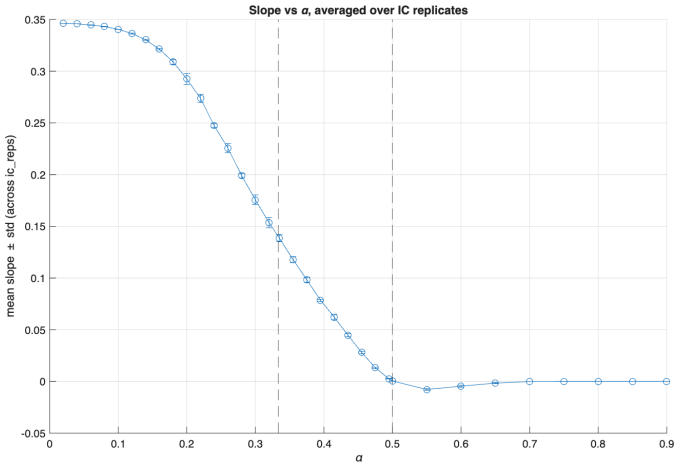


Fig. 7. Mean slope of log-peak versus N, averaged over initial-condition replicates, as a function of α. The feature varies smoothly and monotonically through the open interval, with no jump at a critical α.

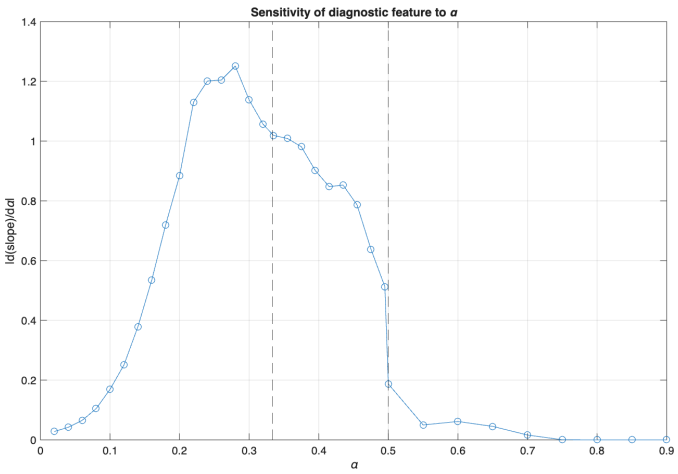


Fig. 8. Feature sensitivity $|d(\text{slope})/d\alpha|$ versus α. Sensitivity peaks near $\alpha \approx 0.25$–0.30, well before the credible-interval-width peak.

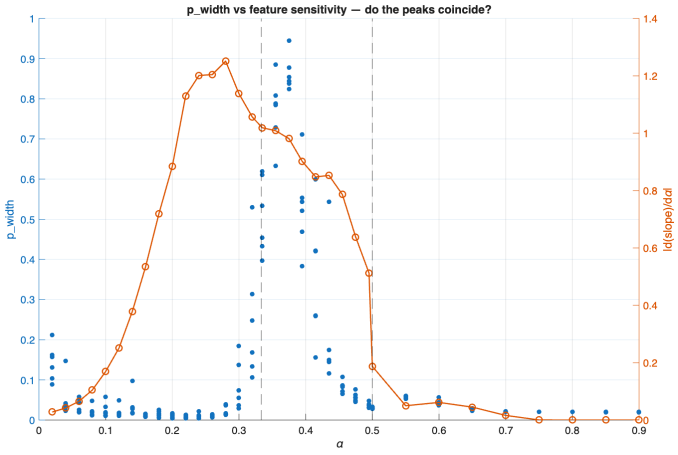


Fig. 9. Credible-interval width (left axis) overlaid with feature sensitivity $|d(\text{slope})/d\alpha|$ (right axis). The peaks are offset, which weakens the explanation that the widening is inherited from where the feature changes fastest.

blow-up boundary (Fig. 9), weakens this specific explanation. It does not exclude others, and it does not show that the widening reflects undecidability rather than another property of the feature set.

Two further explanations remain open. First, any logistic classifier is most uncertain near its decision boundary, and the posterior mean crosses 0.5 inside the open zone (Fig. 2). Second, the widening could reflect intermediate numerical behaviour of the truncated system, whatever the behaviour of the infinite-shell model. A control with a known transition and no open problem was not run, so these explanations are not separated.

Four limitations apply. The study covers one shell model, one feature family, and one classifier architecture. Labels follow the theoretical zone, and the held-out split is by initial-condition replicate, so held-out accuracy measures generalization at the training values of α, not extrapolation. The open zone has no ground truth by construction, so the claim that wide uncertainty there is meaningful is supported only indirectly. Finally, the sampler diagnostics show a well-explored posterior, but a well-mixed sampler on a misspecified model still converges to a confidently wrong posterior. The diagnostics do not show that the features or the logistic model suit the problem.

VII. Conclusion

A Bayesian logistic classifier trained only on the proven zones of a dyadic shell model produced credible intervals 8–13$\times$ wider in the open interval $1/3 \leq \alpha < 1/2$ than in the proven zones. Variation across seeds was small, held-out accuracy was 100%, and predictions stayed confident within 0.01 of both thresholds. The width peak at $\alpha \approx 0.37$–0.38 differed from the feature-sensitivity peak, which weakens one artifact explanation. Decision-boundary and truncation effects were not excluded.

The open regularity problem remains unresolved, and the results do not establish that posterior uncertainty tracks epistemic undecidability. They are an exploratory first step. A control system with a known transition and no open problem is the natural next test.

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